

Bryan M. Maitland, Ph.D.

Research Fish Biologist

Curriculum Vitae

September 2026

Rocky Mountain Research Station,
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Appointments

2024–Present **Research Fish Biologist**, Rocky Mountain Research Station, USDA Forest Service
2023–2024 **Research Associate**, Dept. of Civil and Environmental Engineering, University of Wisconsin-Madison
2022–2023 **Postdoctoral Fellow**, Center for Limnology, University of Wisconsin-Madison
2020–2022 **WI Water Resources Postdoctoral Fellow**, Aquatic Sciences Center (UW-Madison) and WDNR
Affiliate: Graduate Faculty, Dept. of Biology, Geology, and Environmental Science, University of Tennessee at Chattanooga (2026–present).

Education

2020 Ph.D. Ecology University of Wyoming
2015 M.Sc. Conservation Biology (Fisheries) University of Alberta
2011 B.S. Conservation Biology *cum laude* SUNY: College of Environmental Science and Forestry

Research summary

I am a conservation ecologist working on the population and food web ecology of freshwater fishes. My research combines stable isotope ecology, otolith and tissue chemistry, long-term monitoring data, remote sensing, and Bayesian and spatial statistical modeling to ask how climate change, wildfire, altered flow regimes, and land use reshape fish populations and the food webs that support them. I work across systems—headwater streams, large rivers, and the Laurentian Great Lakes—and at scales from individual foraging to continental-scale synthesis. Current work at the Rocky Mountain Research Station centers on salmonids in the interior Columbia River basin: predicting developmental phenology from thermal regimes, mapping spawning activity from satellite imagery, and reconstructing historical baselines to inform co-management of wildfire and aquatic species. A through-line in my work is building open, reusable tools and datasets—R packages, databases, and reproducible workflows—so that the methods transfer to the agencies and partners who must act on the results.

- Since 2014 I have authored 26 peer-reviewed journal articles and 1 R package (3547 downloads), and presented 47 papers and posters at AFS, SFS, ASLO, AGU, IAGLR, and regional meetings.
- Google Scholar h-index 14, 724 total citations (as of 11 September 2026).
- Since 2015 I have led, co-led, or partnered on \$0.3 million in funded projects, and contributed to invited proposals exceeding \$2 million in scope.
- I have reviewed 60 manuscripts for 19 journals since 2016, and served as a grant reviewer for NSF (Division of Environmental Biology) and the Lake Champlain Fisheries Research Program.

Selected publications

Bold indicates my name; underlined names indicate undergraduate research mentees. Citation counts are from Google Scholar on 11 September 2026. A full list of 26 articles is available in my complete CV.

8. Sparks*, M.M., **Maitland*, B.M.**, Felts, E., Swartz, A., & Frater, P. (2026). hatchR: A toolset to predict when fish hatch and emerge. *Fisheries*, 51(2), 49–59. <https://doi.org/10.1093/fshmag/vuaf078> [Citations: 2].
7. Keppeler, F.W., Giarrizzo, T., Montaña, C.G., Adite, A., Akın, Şenol, Angelini, R., **Maitland, B.M.**, Arantes, C.C., Benedito, E., Bokhutlo, T., El-Sabaawi, R., Garcia, A., Gonzalez-Bergonzoni, I., Hoeinghaus, D., Hoffman, J., ... Winemiller, K.O. (2026). What determines trophic niche breadth? A global analysis of freshwater fishes using isospaces. *Reviews in Fish Biology and Fisheries*, 36(13). <https://doi.org/10.1007/s11160-025-10023-2> [Citations: 2].
6. **Maitland, B.M.**, Bootsma, H.A., Bronte, C.R., Bunnell, D.B., Feiner, Z.S., Fenske, K.H., Fetzner, W.W., Foley, C.J., Gerig, B.S., Happel, A., Höök, T.O., Keppeler, F.W., Kornis, M.S., Lepak, R.F., McNaught, A.S., ... Jensen, O.P. (2024). Testing food web theory in a large lake: The role of body size in habitat coupling in Lake Michigan. *Ecology*, 105(10), e4413. <https://doi.org/10.1002/ecy.4413> [Citations: 14].
5. Barrus, N.T., **Maitland, B.M.**, & Rahel, F.J. (2024). Assessing a standardized method to identify optimal baselines for trophic position estimation in stable isotope studies of stream ecosystems. *Hydrobiologia*, 851, 4673–4691. <https://doi.org/10.1007/s10750-024-05618-y> [Citations: 3].
4. **Maitland, B.M.**, & Rahel, F.J. (2023). Aquatic food web expansion and trophic redundancy along the Rocky Mountain–Great Plains ecotone. *Ecology*, 104(7), e4103. <https://doi.org/10.1002/ecy.4103> [Citations: 13].
3. **Maitland, B.M.**, & Latzka, A.W. (2022). Shifting climate conditions affect recruitment in Midwestern stream trout, but depend on seasonal and spatial context. *Ecosphere*, 13(12), e4308. <https://doi.org/10.1002/ecs2.4308> [Citations: 17].

2. **Maitland, B.M.**, & Rahel, F.J. (2021). Nonlethal fin sampling of North American freshwater fishes for food web studies using stable isotopes. *North American Journal of Fisheries Management*, 41(2), 410–420. <https://doi.org/10.1002/nafm.10539> [Citations: 18].
1. Craig, L.S., Olden, J.D., Arthington, A.H., Entekin, S., Hawkins, C.P., Kelly, J.J., Kennedy, T.A., **Maitland, B.M.**, Rosi, E.J., Roy, A.H., Strayer, D.L., Tank, J.L., West, A.O., & Wooten, M.S. (2017). Meeting the challenge of interacting threats in freshwater ecosystems: A call to scientists and managers. *Elementa: Science of the Anthropocene*, 5(72), 1–15. <https://doi.org/10.1525/elementa.256> [Citations: 205].

Selected funding

2026–2028	Co-PI. “From flames to fish: Co-management for wildfire and aquatic species in Idaho”. <i>Joint Fire Science Program</i> .	\$150,284
2025–2027	Lead-PI. “Monitoring Chinook Salmon spawning from space”. <i>Trout Unlimited</i> .	\$38,122
2024–2027	Lead-PI. “Chinook Salmon life history and otolith microchemistry”. <i>UC-Davis and USDA Forest Service</i> .	\$50,184
Pending	Co-PI. “Idaho Bull Trout migration timing and habitat use”. <i>U.S. Fish and Wildlife Service</i> .	\$159,900
Pending	Lead-PI. “From reservoir to redd: linking reservoir operations to bull trout demography”. <i>U.S. Bureau of Reclamation</i> .	\$145,000
Pending	Lead-PI. “Idaho Bull Trout food web ecology and hybrid interactions”. <i>U.S. Fish and Wildlife Service</i> .	\$20,000
2019–2020	Lead-PI. “Diet specialization along stream gradients”. <i>UWyo Zoology</i> .	\$22,398

Selected awards and honors

2026	Outstanding Service Award, USDA Forest Service, Rocky Mountain Research Station
2024	Outstanding Service Award, USDA Forest Service, Rocky Mountain Research Station
2021	Best Professional Presentation Award, Wisconsin Chapter American Fisheries Society
2020	Wisconsin Water Resources Postdoctoral Fellow, University of Wisconsin-Madison
2019	Dean’s Graduate Fellowship, College of Arts and Sciences, University of Wyoming
2018	Eugene Maughan Graduate Award, Western Division American Fisheries Society

Mentoring and teaching

2023–present	Supervised 3 graduate/post-baccalaureate researchers (U. Idaho, UW-Madison, UT-Chattanooga); MS committee member, UT-Chattanooga
2015–2025	Mentored 11 undergraduate researchers, two of whom have led or co-authored peer-reviewed publications
2014–2020	Instructor of record for Ichthyology (UWyo); teaching assistant for 12 sections of Ichthyology and Fisheries Management (UWyo, U. Alberta)

Selected service and synergistic activities

2025–2029	Mentor of Practice, NSF IRES: Climate Change Resilience in an Aquatic Bright Spot (Jensen, PI, UW-Madison)
2026	Program co-chair, <i>Fish and Fire</i> symposium, American Fisheries Society Annual Meeting
2024	Program co-chair, <i>Science to Restore and Manage Wildland Aquatic Habitats and Watersheds</i> , AGU Fall Meeting
2020–2022	Member, Trout Species Team, Wisconsin Dept. of Natural Resources; Fisheries Working Group, Wisconsin Initiative on Climate Change Impacts
2015–present	Member, Freshwater Working Group, Society for Conservation Biology; AFS, SCB, ESA, SFS

Developer and maintainer of *hatchR*, an open-source R package for predicting fish hatch and emergence timing from thermal regimes, with accompanying workshops delivered to state and federal fisheries staff. Research covered by regional and national media outlets.